

V Madhan Kumar

Bengaluru, Karnataka, India · Available immediately

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🌐 LinkedIn 🐙 GitHub 📁 Portfolio

Education

National Institute of Technology, Karnataka, Surathkal

2019 – 2023

BTech in Computer Science Engineering

CGPA: 8.47

Minors in Management

CGPA: 8.6

Experience

Jumbo (JoinJumbo), Delhi — *Backend Engineer*

Sep 2025 – Jun 2026

Primary contributor across 12 backend services — Node.js/TypeScript, Go, AWS Lambda, PostgreSQL, Redis

- **Backend Microservices at Scale** — Shipped across Lambda microservices and a NestJS monolith — read-heavy APIs, async job processing, and transactional wallet flows on a high-traffic consumer platform.
- **Caching & Performance** — Redis-first + DB fallback with granular invalidation, cache warm paths, ElasticCache TLS migration, and CDN edge APIs — p99 from 128ms to sub-30ms; 70–90% DB query reduction on hot paths.
- **Async & Stream Processing** — Redis Stream → Go consumer for burst enrollment writes — capacity pre-validation, wallet-before-lock ordering, DLQ depth alerts, and SQS/EventBridge async jobs.
- **Ranking, Metrics & Data Layer** — User metrics aggregation APIs and batch rank computation via ClickHouse; Aurora PostgreSQL tuning — strategic indexes, query refactors (EXISTS vs full-row loads), parameter tuning, and CloudWatch Performance Insights.
- **Transactional Systems & Reliability** — Idempotent webhook ingestion, multi-provider wallet credit/debit with atomic reversal on failed settlement; Testcontainers integration test CI, P0 concurrency hardening, and internal admin APIs.

PlayGPL (Litifer), Bengaluru — *Software Developer*

May 2023 – Mar 2025

- **Backend APIs & Microservices** — REST APIs using Golang and Node.js with authentication, authorization, rate limiting, and analytics for a high-traffic consumer platform.
- **Fraud Detection Pipeline** — Real-time fraud prevention using device fingerprints, network behavior analysis, and root/VPN checks to block referral abuse and malicious transactions.
- **Platform Integrations** — Payment and notification integrations, referral/reward flows, and operational dashboards supporting day-to-day product releases.

Projects

SQL Query Analyzer & Optimizer

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FastAPI, sqlglot, Claude API, PostgreSQL, Docker

- Parses SQL via sqlglot AST traversal, detects anti-patterns (SELECT *, missing WHERE, excessive joins), and scores query complexity.
- Integrated Claude API for index/rewrite suggestions with rule-based fallback, data lineage tracking, and paginated query history — extends production Aurora tuning work.

Technical Skills

- **Languages:** TypeScript, Golang, Node.js, Python, SQL, JavaScript
- **Backend:** NestJS, Express, FastAPI, REST APIs, Microservices, Distributed Systems, System Design
- **Data:** PostgreSQL (Aurora), Redis, ClickHouse, TypeORM, query optimization, batch analytics
- **Cloud:** AWS (Lambda, SQS, EventBridge, Aurora/RDS, ElasticCache, API Gateway, S3, IAM), Docker
- **Practices:** Git, GitHub Actions, Testcontainers, integration testing, idempotency patterns, CloudWatch, CI/CD